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Prediction of open-pit mine truck travel time based on LSTM-TabTransformer

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Truck travel time prediction is the basis for real-time optimal scheduling decision of open-pit trucks. Most prediction models for open-pit truck travel time mainly rely on a single machine learning algorithm or model to optimize the hyperparameters, which is difficult to accurately capture the composite characteristics of truck travel time data, affectting the accuracy of the prediction results. This paper proposed a travel time prediction method for open-pit trucks based on LSTM-TabTransformer. After eliminating the outliers in the truck travel time data set by applying the Paura criterion, the data set was non-dimensionalized, and the corresponding category data was processed by means of data diversion. Then, the self-attention mechanism of TabTransformer and gating mechanism of LSTM were used to capture the dynamic characteristics of the travel time variation rule at multiple levels. Finally, the dimensionality of the concatenated feature matrix was reduced by MLP, and the predicted truck travel time was output. Based on the truck travel time series data set collected from the example open-pit coal mine, the prediction experiment was carried out. The results show that the combined machine learning model LSTM-Tabtransformer proposed in this paper is suitable to process the complex truck travel time series data set and learn multiple features, which can reduce the volatility of relative absolute error (RAE) and relative square error (RSE) of travel time prediction, overcome the limitations of a single machine learning prediction model, effectively increase the sensitivity of learning features, and significantly improve the accuracy of truck travel time prediction in open-pit mines.

Keywords Open-pit mine, Truck optimal dispatch decision, Truck travel time prediction, LSTM-TabTransformer, Combined machine learning model

According to the data from the Ministry of Natural Resources of PRC, by the end of 2022, there were 357 open-pit coal mines in China, with a total production capacity of 1.162 billion tons. The average annual production capacity of a single open pit coal mine was 3.25 million tons, which was approximately three times that of an underground mine, and there were more than 30 open-pit coal mines with an annual production capacity of 10 million tons[1]. Open pit coal mine production capacity accounts for about 23% of China's total coal output. The single bucket-truck intermittent mining technology is flexible and adaptable, it has been widely adopted in open-pit coal mines in China since the 1980s[2]. In the single bucket-truck intermittent mining system, the investment in truck transportation system accounts for about 40% ~ 60% of the total investment in the mine, and the transportation cost accounts for about 40% ~ 70% of the total production cost of the coal. According to statistics, by optimizing the truck scheduling scheme, the truck transportation cost can be reduced by 3% ~ 20%, and the economic benefit is remarkable[3]. In the process of truck scheduling scheme optimization, the truck travel time prediction is a vital decision-making basis.

To accurately predict the travel time of trucks in open-pit mines, Qingshan Sun[4] established a statistical mathematical model for multiple types of trucks, and the travel time of different types of trucks on different routes was determined by statistical methods. Runcai Bai[5], Jiangang Li[6], and other researchers proposed open-pit mine truck travel time prediction models based on BP (Back Propagation Neural Network) and ANFIS (Adaptive Network-based Fuzzy Inference System), they made a meaningful attempt in the application of artificial intelligence methods to predict the truck trval time in open pit mine. Xue et al.[7] introduced a selective ensemble learning algorithm based on least-squares support vector regression to dynamically predict truck travel time, and achieved satisfactory prediction results on the experimental data collected from open-pit coal mines.

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Xiaoyu Sun et al.[8] compared and analyzed the performance of machine learning algorithms such as k-Nearest Neighbors (kNN), Support Vector Machine (SVM), and Random Forest (RF) in open-pit mine truck travel time prediction, and found that the prediction accuracy of machine learning algorithms was 15.79% higher than that of traditional mean algorithms, and that SVM and RF algorithms performed better than kNN. Qinghua Gu et al.[9,10] applied the SVM, Adaboost, RF, and HGSVMA models to predict truck travel time in open-pit mines. Their experimental results showed that the MSE and MAE values of the HGSVMA model were smaller than SVM and RF, while the R^2 values were higher. The MSE and MAE values were reduced by 76.96% and 29.90% respectively, and the R^2 value increased by 4.48%, indicating that the ensemble learning model outperformed the single model in predicting truck travel time in open pit mines.

The existing research results mainly apply traditional statistical methods, machine learning methods, deep learning methods, and ensemble learning methods. In particular, the current ensemble learning methods lack research on time-series features. Although the machine learning methods have been applied in predicting truck travel time in open-pit mines[11–13], in the actual production environment of open-pit mines, the travel data of trucks are time-series data with nonlinear characteristics, and their variations are influenced by various factors (such as material properties, road quality, weather conditions, working period, truck types, etc.), a single prediction model which relies on fixed length time series features is difficult to effectively capture the composite chacteristics in the truck travel time data set, thus affecting the prediction accuracy. Based on the above research background, this paper introduces the TabTransformer model[14], and combines it with the Long Short-Term Memory (LSTM)[15] to construct a LSTM-TabTransformer hybrid prediction neural network model for open-pit mine truck travel time. Taking advantages of the self-attention and long short-term memory mechanism, LSTM-TabTransformer model can directly capture the dependencies between different positions in the data sequence and effectively avoid the problems of gradient vanishing and explosion during the training. Experiment results show that the proposed LSTM-Tabtransformer model can significantly improve the prediction accuracy of open-pit mine truck travel time, which is of great significance for optimizing the open-pit mine truck scheduling scheme, improving the truck operation efficiency and utilization, and reducing the production cost.

Influencing factors of truck travel time in open-pit mine

The truck travel time of open-pit mine is mainly affected by the following factors :

(1) Material properties. The materials loaded by trucks in open-pit mines are mainly stripped soil, rock, or mined ore. The different block size, unit weight and unloading position of the loaded materials will affect the actual load, travel path, speed, unit fuel consumption and other indicators of the truck, and ultimately affect the travel time and cost of the truck from the loading position to the unloading position.

(2) Road quality. Road grade, flatness, slope, plane curve radius, vertical curve radius, etc., will have different degrees of impact on truck travel time.

(3) Weather conditions. Rainfall, snowfall, gale, dust, dense fog, etc., will lead to adverse driving conditions such as reduced visibility, wet and slippery road face, ponding on the road, which will reduce the travel speed and prolong the travel time of trucks.

(4) Working period. Open-pit mines generally adopt the 24-hour continuous working system. When the truck working period is different, the travel time on the same road section will also be greatly different due to the influence of line of sight conditions.

(5) Type of the truck. An open-pit mine is usually equipped with different types of trucks according to the property of the transported materials. The truck load capacity, engine power, service time, etc., will determine the speed and fuel consumption of the truck, thus affecting the travel time and cost of the truck.

(6) Travel distance. Truck travel distance is the key factor to determine the truck travel time. In general, the truck travel time will increase with the travel distance.

(7) Driver quality. Including the driver's proficiency in truck control, familiarity with road conditions, mental state, etc.

Besides the above, road traffic density, cross transportation, temporary road control, truck load status, etc., will also have an impact on truck travel time.

Among the infulencing factors, some are expressed directly in numerical form, and some are expressed in text, symbols and other forms.

LSTM-TabTransformer: A combined model for truck travel time prediction in open-pit mine

The prediction of truck travel time is a complex nonlinear correlation problem. The truck travel time is affected by many deterministic, random, quantifiable and non-quantifiable factors, and it is difficult to be explicitly described by traditional mathematical methods. Referring to the idea of applying artificial intelligence methods to predict the truck travel time of open-pit mines in previous research results, this paper proposes a combined LSTM-TabTransformer truck travel time prediction model based on machine learning.

TabTransformer

Truck travel time prediction in open-pit mines is affected by equipment capacity, road quality, weather conditions, working period and other factors. Due to different data characteristics of these factors, the training data set of the truck travel time prediction model includes category, number, text, time and other types of data information. Taking each training data as a row and each influencing factor as a column, the open-pit mine truck travel time prediction data set belongs to typical tabular data.

In the field of machine learning, methods such as Logistic Regression (LR), Decision Trees (DT), Random Forests (RF), Support Vector Machines (SVM), Multi-layer Perceptron (MLP), and Gradient Boosted Decision

Trees (GBDT)[16–18] are commonly used for tabular data modeling. Among those methods, GBDT and MLP are considered advanced. GBDT offers fast training, ease of interpretation, and competitive prediction accuracy, but it has several shortcomings: (1) It is difficult to adapt to the continuous training with streaming data, and it is also unable to perform the effective end-to-end learning on multimodal tabular data; (2) It is not suitable for the most advanced semi-supervised learning method; (3) It cannot process tabular datasets containing missing or noisy data, and its robustness has not been fully verified. In contrast, MLP uses a gradient descent strategy for model training, enabling it to carry out end-to-end learning on multimodal tabular data. However, due to its shallow structure and context-free embedded coding features, MLP is not robust enough for tabular data containing missing and noise, and has not achieved satisfactory performance for semi-supervised learning.

In 2020, Xin Huang et al.[14] proposed a deep neural network model architecture TabTransformer for tabular data processing based on Transformer, which better solved the limitations of MLP and other deep learning models, and also bridged the performance gap between MLP and GBDT. The TabTransformer model architecture is shown in Fig. 1.

Suppose (x, y) is a feature-target data pair, where $x \equiv \{x_{cat}, x_{cont}\}, x_{cat} \equiv \{x_1, x_2, \cdots, x_m\}$ is a category feature (here, it represents all discrete features, i.e., features that can be represented by natural numbers), $x_{cont} \in R^c$ represents all continuous features. The data processing process of TabTransformer is as follows: Firstly, each category feature is embedded into a parameter embedding with dimension d by column embedding. Then, the parameter embedding results are input into multiple Transformer layers in turn (the output of the previous Transformer layer is used as the input of the next Transformer layer). When passing through the last Transformer layer, all parameter embedding results are transformed into context embedding by context continuous aggregation. Finally, the $d \times m + c$ dimension vector composed of context embedding and continuous features is input into MLP, and the prediction result y is output.

The TabTransformer loss function is :

$$L(x, y) = H\left(g_{\psi}\left(f_{\theta}\left(E_{\varphi}\left(x_{cat}\right)\right), x_{cont}\right), y\right) \tag{1}$$

Where, $E_{\Phi}(x_{cat})$ is the parameterization embedding result, f_{θ} is the Transformer layer sequence, and g_{ψ} is the multi-layer perceptron MLP.

TabTransformer takes the cross entropy as the loss function of classification task, and the mean square error as the loss function of regression task. In the end-to-end learning, the loss function is minimized by a gradient descent method to determine the TabTransformer parameters.

LSTM-TabTransformer

In order to make TabTransformer better adapt to the task of truck travel time prediction in open-pit mines, this paper improves the TabTransformer by including adding position coding to the feature information end, using LSTM to extract continuous time series features, and applying cosine annealing attenuation learning rate. The improved LSTM-TabTransformer architecture is shown in Fig. 2.

Position encoding of feature information

In order to enable the TabTransformer model to distinguish the relative positions between different features, position encoding is added in the process of embedding and encoding feature information. On the one hand, the discrete position information is expressed as a continuous numerical vector. On the other hand, by adding position encoding, the model can perceive the order and relative position of different features in the input data, so as to better capture the sequence information and more accurately predict the travel time of open-pit trucks.

The calculation formula for position encoding is shown in Eq. (2), and the visualization of the position encoding is shown in Fig. 3.

$$PE_t^{(i)} = \begin{cases} \sin\left(\frac{t}{n^{\frac{2k_{map}}{d_{sim}}}}\right), & i = 2k \\ \cos\left(\frac{t}{n^{\frac{2k_{map}}{d_{sim}}}}\right), & i = 2k+1 \end{cases} \tag{2}$$

Where, $PE_t^{(i)}$is the position function used to map the element at position k_{map} in the input sequence to the position matrix; t is the position of the object in the input sequence; d_{sim} is the dimension of the output embedding layer; $k_{map}$is the mapped column index.

Due to the properties of the sine and cosine functions, each value in the position vector lies between [−1, 1]. At the same time, from a vertical perspective, the positions at the rear of the position encoding diagram have smaller frequencies and longer wavelengths, and changes of t have a minimal impact on the final position encoding.

Continuous time-series feature extraction based on LSTM

Long Short-Term Memory (LSTM) is a variant of Recurrent Neural Network (RNN), which is specifically used to solve the problem of gradient disappearance and explosion in traditional RNN. LSTM can not only effectively manage and process historical information, but also introduce the memory unit of 'Cell State', which can better deal with long-term dependence problems and improve the learning ability of the model for time-series features [19–20]. In this paper, LSTM is applied to extract continuous time-series features in truck travel time prediction, and its core gating structure is shown in Fig. 4.

In LSTM, the hidden state of each time step is controlled by three gating units, namely input gate, forgetting gate and output gate.

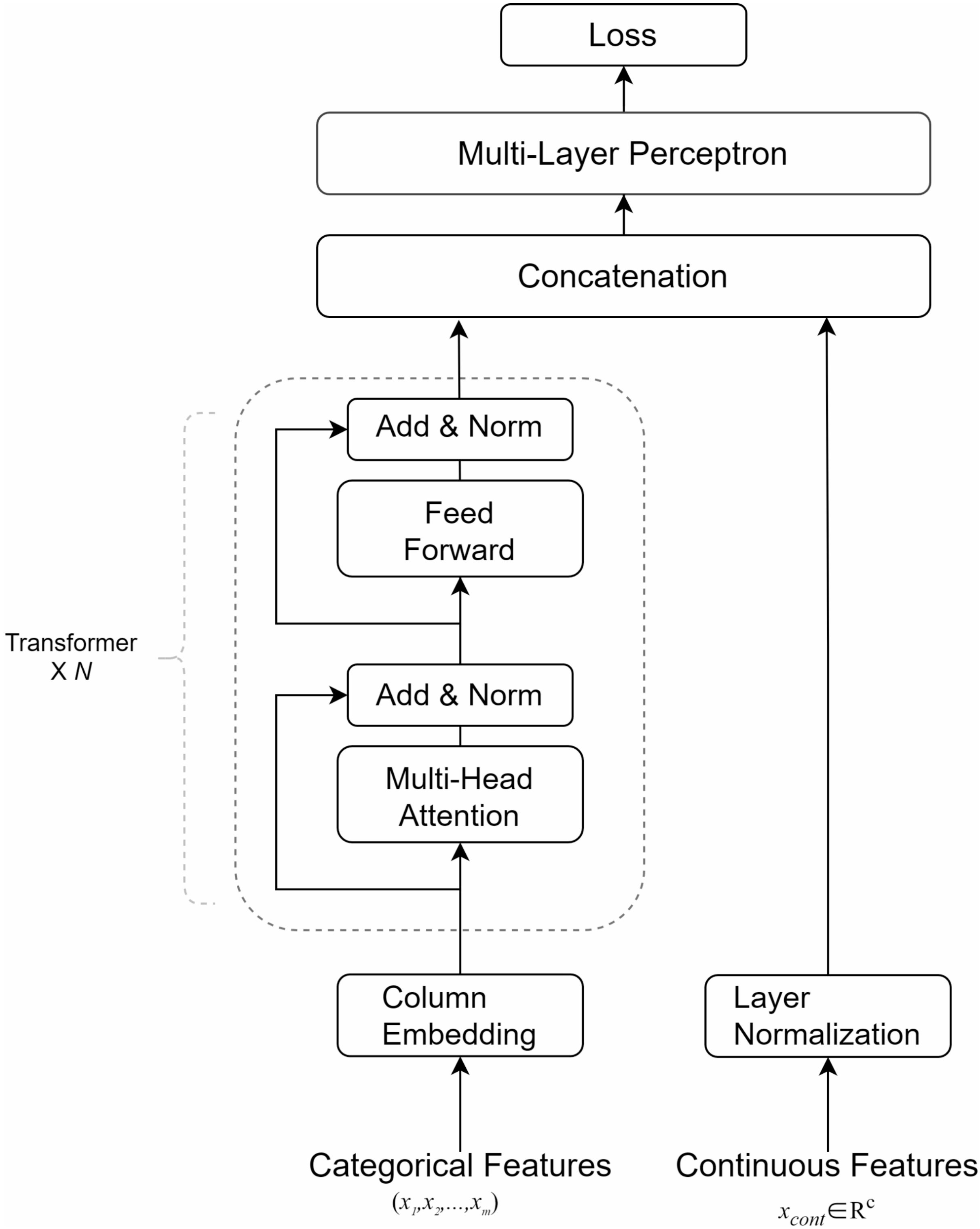

Fig. 1. TabTransformer architecture diagram.

The time series feature data such as speed and position information of each time step are used as the input of LSTM. The LSTM model will update and calculate a new vector i_t according to the previous state h_{t-1} of the time series feature information and the current input. After that, based on the previous state h_{t-1}, the input data will be passed through the forgetting gate and a new vector f_t will be calculated to represent the old open-pit mine truck travel information which needs to be retained. Finally, the cell state of the current node is calculated. By concatenating the previous state C_t with the current node input and calculating the output h_{t-1}, LSTM will determine which information related to the travel characteristics is output through the output gate, and calculate

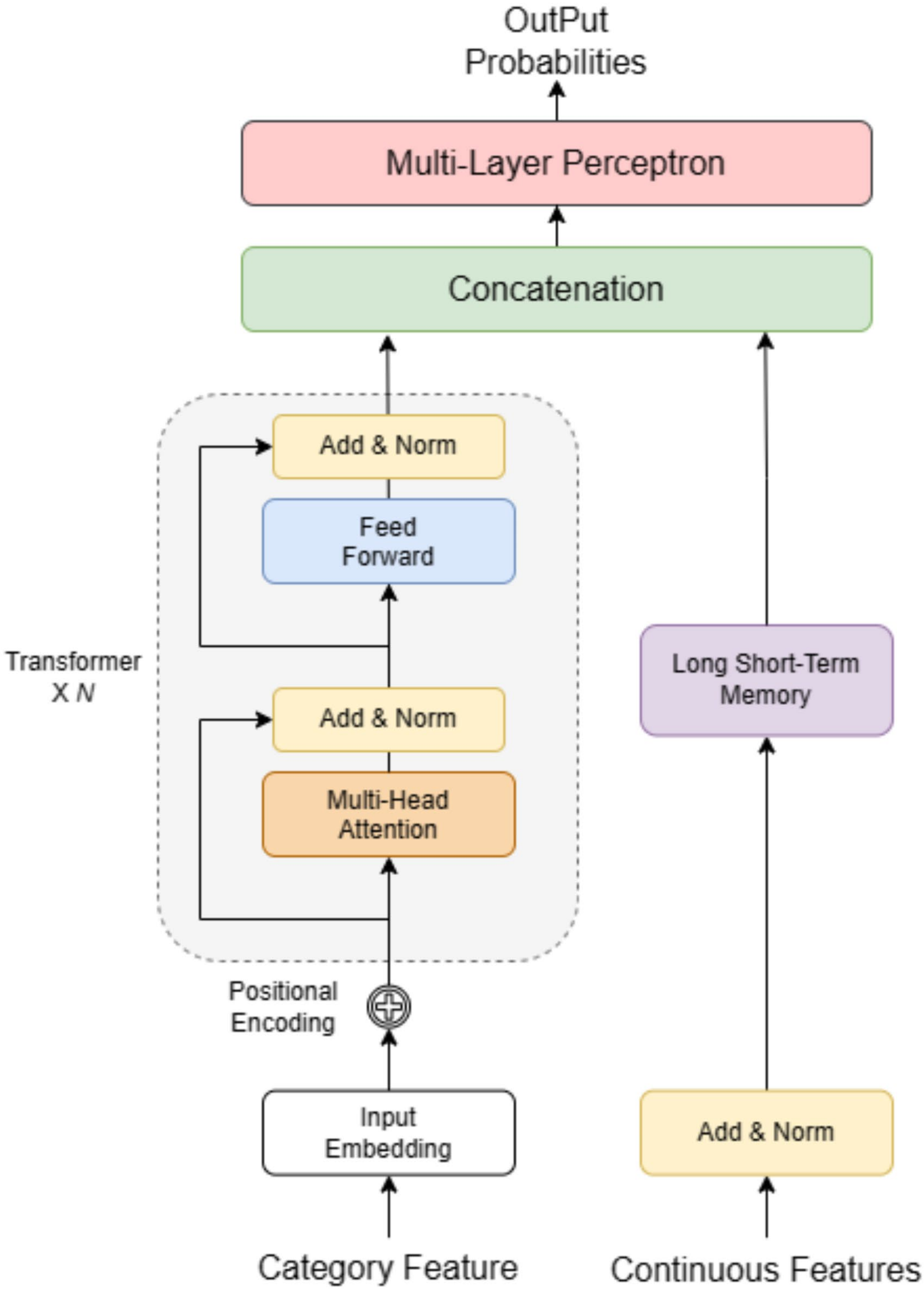

Fig. 2. LSTM-TabTransformer architecture diagram.

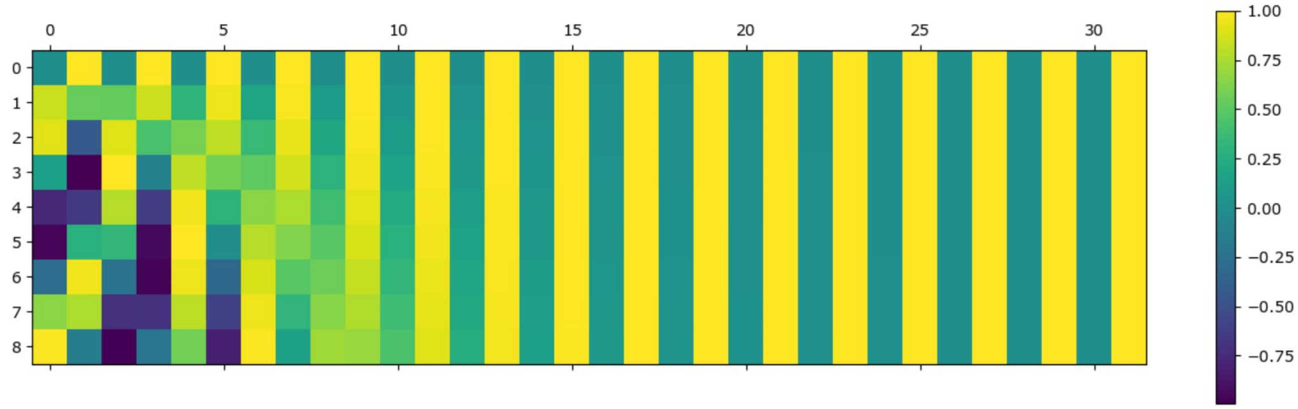

Fig. 3. Feature information position encoding.

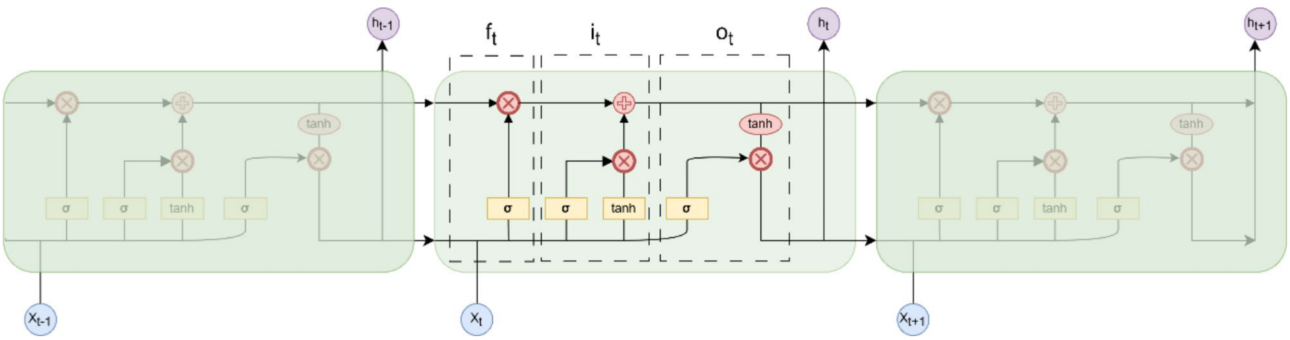

Fig. 4. LSTM gating structure.

a new state h_t by combining the current cell state C_t with the output o_t. The calculation formula of each step is as follows (3):

$$\begin{cases} i_t = \sigma\left(W_i \cdot \left[h_{t-1}, x_t\right] + b_i\right) \\ f_t = \sigma\left(W_f \cdot \left[h_{t-1}, x_t\right] + b_f\right) \\ C_t = f_t \cdot C_{t-1} + i_t \cdot \widetilde{C}_t \\ \widetilde{C}_t = \tanh(W_C \cdot \left[h_{t-1}, x_t\right] + b_C) \\ o_t = \sigma\left(W_o \cdot \left[h_{t-1}, x_t\right] + b_o\right) \\ h_t = o_t \cdot \tanh(C_t) \end{cases} \tag{3}$$

Where, $\sigma(X)$is the ReLu activation function, f_t is the forgetting gate, i_t is the input gate, o_t is the output gate, $\widetilde{C}_t$ is the candidate cell state, C_t is the cell state at the current time step, h_t is the hidden state of the current time step, *W* and *b* are the weight matrix and bias term, respectively.

The final time sequence feature output is represented by $U_{continu}$, and its specific structure is shown in formula (4).

$$U_{continu} = [Batch\_s, h_t] \tag{4}$$

Where, *Batch_s* represents the continuous time-series features extracted by the LSTM.

Learning rate attenuation based on cosine annealing

In the process of deep learning, when getting closer to the global minimum of the Loss value, the model cannot continue to fit without changing the learning rate. In order to better optimize the training process and improve the performance of the model, this paper introduces the cosine annealing method to reduce the learning rate, so as to balance the global search and local convergence, improve the stability of the training, avoid falling into the local optimal solution, and accelerate the convergence speed.

Cosine annealing learning rate update formula is:

$$\eta_t = \begin{cases} \eta^i_{min} + \frac{1}{2}(\eta^i_{max} - \eta^i_{min})(1 + cos(\frac{T_i}{T_{max}}\pi)), & T_i < T_{max} \\ \eta^i_{min} + \frac{1}{2}(\eta^i_{max} - \eta^i_{min})(1 + cos(1 - \frac{T_i}{T_{max}}\pi)), & T_{max} < T_i < 2T_{max} \end{cases} \tag{5}$$

Where, η_t is the learning rate, η^i_{max} and η^i_{min} have maximum learning rate and minimum learning rate respectively, T_i is the current number of iterations, T_{max} is the maximum number of iterations.

The change trend of learning rate is shown in Fig. 5.

This paper introduces LSTM to extract the continuous data features from truck travel time data. Instead of improving the classical LSTM, a hybrid model is constructed by combining the classical LSTM with TabTransformer to solve the complex nonlinear correlation problems.

Experiment and analysis

Data sources

The basic data of truck travel time prediction in this paper are collected from an open-pit coal mine in Inner Mongolia, China. The geographical location is shown in Fig. 6, the distribution of mining area is shown in Fig. 7, and the structure of truck transportation road network is shown in Fig. 8. The example open-pit coal mine adopts comprehensive mining technology, and the total annual mining and stripping amount is about 180 million m^3, of which about 150 million m^3 of coal and rock needs to be transported by trucks to the unloading position.

In this paper, a total of 200,000 training data sets of the prediction model are extracted from the example open-pit mine truck dispatching system[14] from December 2022 to July 2023. Each data record is composed of truck number, shovel number, truck type, actual loading time, actual travel time, position coordinates, coal and rock category and other information. In order to incorporate the weather conditions into the influencing factors of truck travel time, the single-day meteorological data extracted from No.53553 station of China National Meteorological Science Data Center are also included. After data processing, 10,000 pieces of truck actual travel time sample data are divided into training set and validation set according to 8 : 2.

The quality and the type of training data determine the upper limit of the model effect, and the goal of model training is to approach this upper limit as close as possible. The influence of data on the prediction model training is very important, which will directly affect the prediction accuracy and generalization ability of the model. The input and output variables and data selected in this paper are shown in Table 1.

Shovel serial number, truck serial number, shovel type, truck type, material property, weather conditions, road quality, working period, lookout conditions are classified according to the type of *n*, and the numbers *1*, *2*, *3*,…, *n* are used to input the model.

Data pre-processing

The truck travel data collected from the example open-pit coal mine come from different monitoring instruments. Due to the interference of various complex factors in the on-site production environment, there are some abnormal data in the data set that are significantly different from the real value. These abnormal data will affect the training effect of the prediction model and its later application. In this paper, the Pauta criterion is used to clean the truck travel time data set to eliminate abnormal data and improve the model training effect

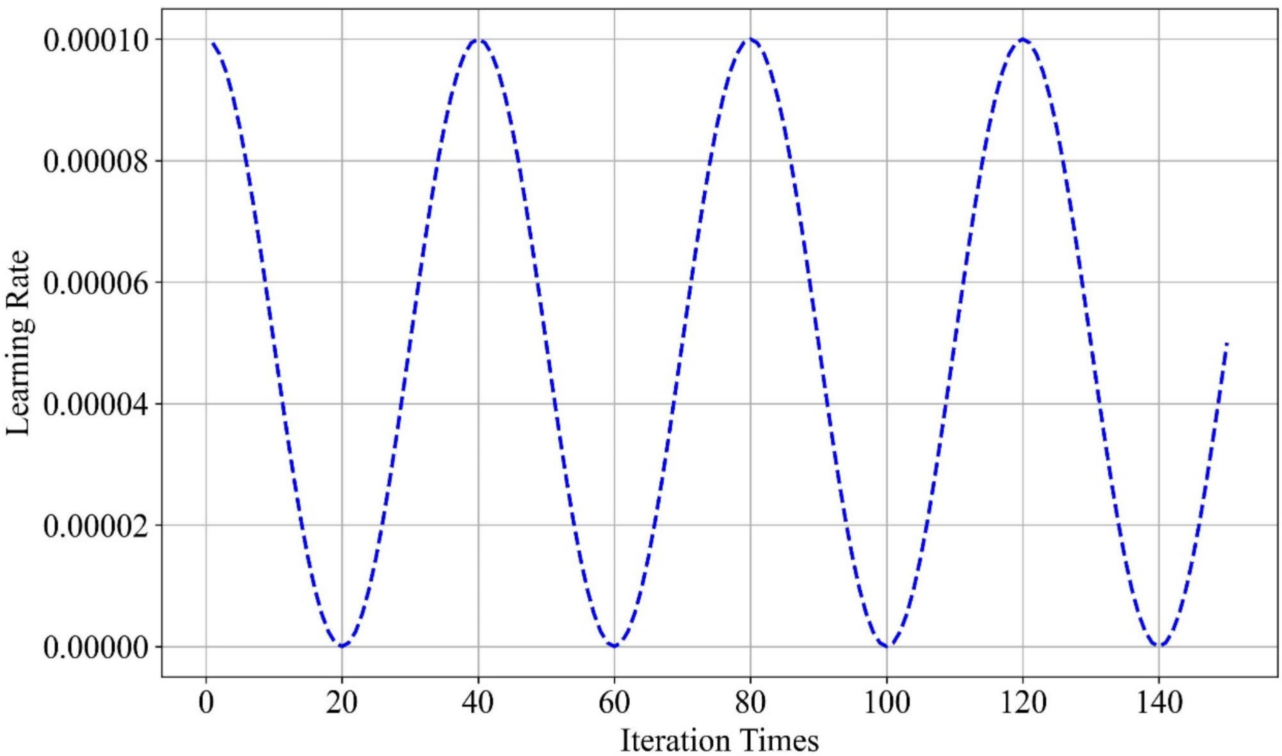

Fig. 5. Cosine annealing learning rate curve.

Fig. 6. Geographic location map of example open pit coal mine in Inner Mongolia, China (location - Google Earth).

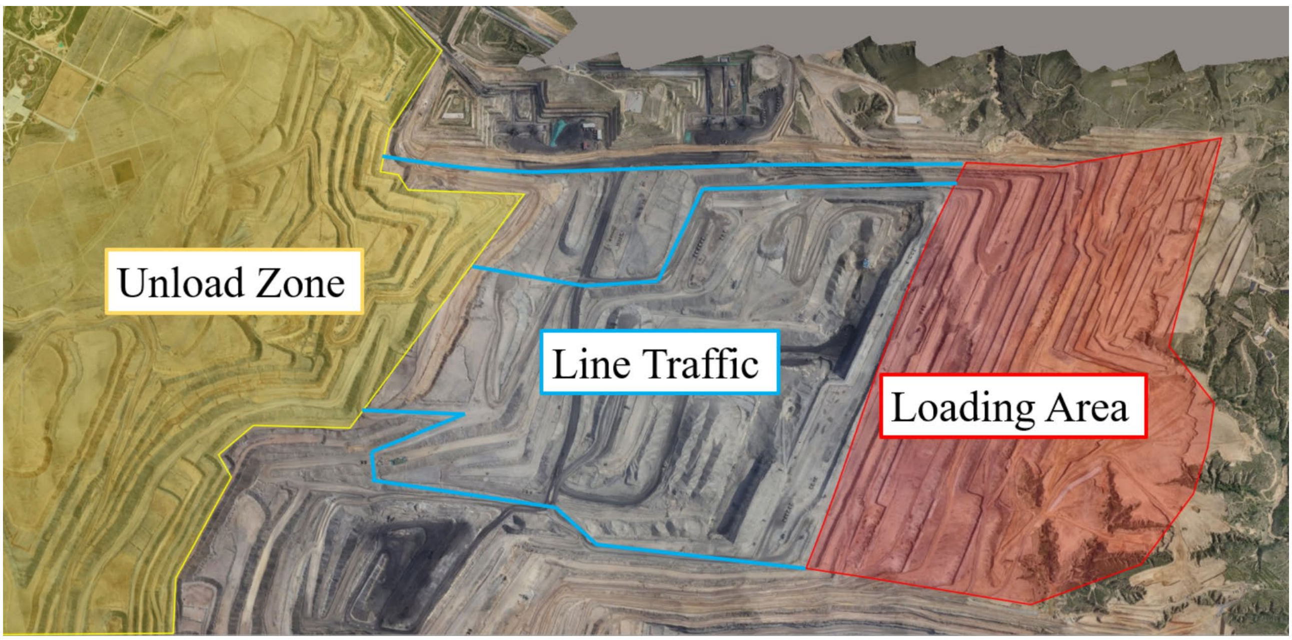

Fig. 7. Actual road network of example open pit coal mine by Acute3D Viewer (The basic map is generated from the point cloud of the mining area).

and prediction accuracy. Assuming that the index is measured with equal precision, $x_1, x_2 \cdots x_n$ is obtained independently, and its arithmetic mean x and residual error $v_i = x_i - x(i = 1{,}2, \cdots, n)$ are calculated, and the standard deviation σ is calculated according to Bessel formula (6). If the residual error v_b $(1 \leq b \leq n)$ of a measured value x_b satisfies Eq. (7), x_b is considered to be a bad value with gross error value and should be eliminated.

$$S = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \overline{x})^2}{n-1}} = \sqrt{\frac{(x_1 - \overline{x})^2 + (x_2 - \overline{x})^2 + \cdots + (x_n - \overline{x})^2}{n-1}} \tag{6}$$

$$|v_b| = |x_b - x| > 3\sigma \tag{7}$$

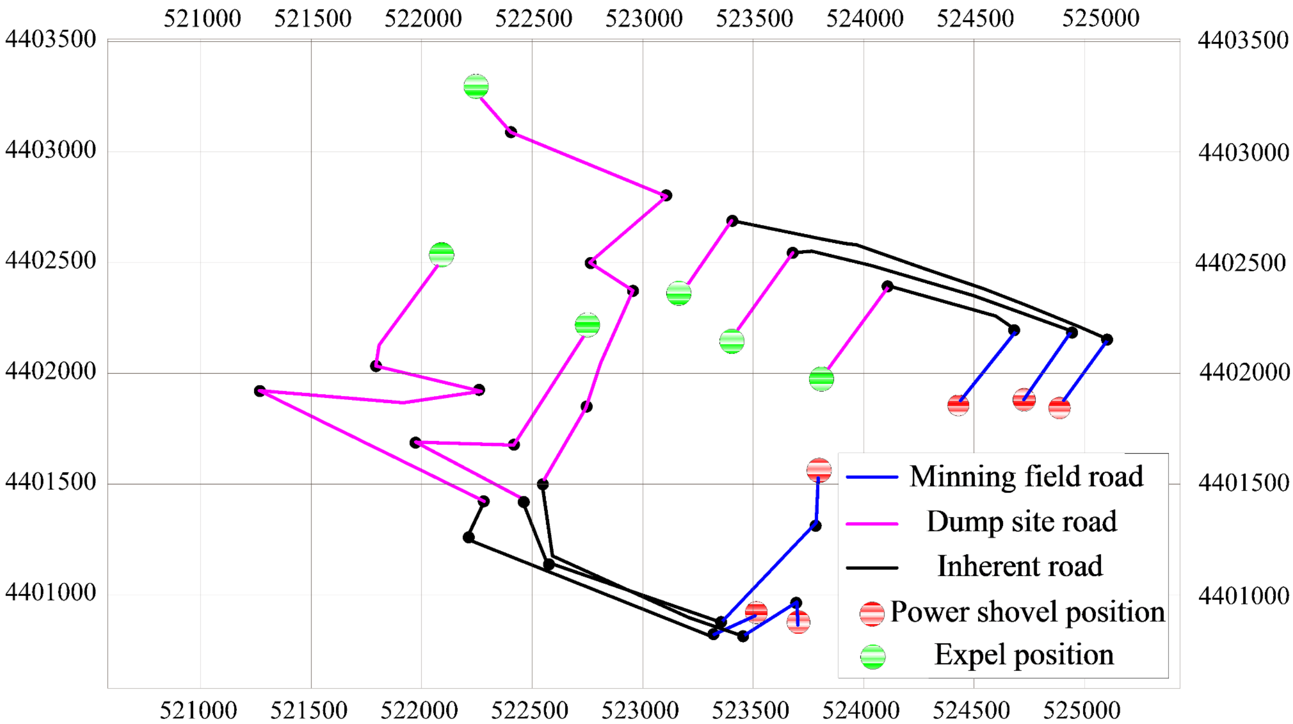

Fig. 8. Road network diagram of example open pit coal mine by 3DMine.

Variable Name	Data Type	Data Role	Data Description	Data Example
Electric shovel serial number	Category	Feature Variable	Sequence number of the shovel	12,604
Truck serial number	Category	Feature Variable	Sequence number of the truck	11,502
Electric shovel type	Category	Feature Variable	Type of the shovel	黑4#WK55
Truck type	Category	Feature Variable	Type of the truck	黑502
Material property	Category	Feature Variable	Type of the material	Rock
Truck load	Continuity	Feature Variable	Actual load of the truck	105(t)
Shovel starting position longitude coordinate	Continuity	Feature Variable	Longitude coordinates of the shovel starting position	156.22°/E
Shovel starting position latitude coordinates	Continuity	Feature Variable	Latitude coordinates of the shovel starting position	48.33°/N
Unloading position longitude coordinates	Continuity	Feature Variable	Longitude coordinates of the unloading position	156.82°/E
Unloading postition latitude coordinate	Continuity	Feature Variable	Latitude coordinates of unloading position	48.55°/N
Shovel end position longitude coordinates	Continuity	Feature Variable	Longitude coordinates of the shovle end position	156.22°/E
Shovel end position latitude coordinates	Continuity	Feature Variable	Latitude coordinates of the shovel end position	48.44°/N
Weather conditions	Category	Feature Variable	Basic meteorological data	Cloudy ~ Light rain
Road quality	Category	Feature Variable	Truck's different speed limits on different road section	Main-line
Working period	Category	Feature Variable	Working period of the truck	Night shift
Lookout conditions	Category	Feature Variable	Driver's visual distance	Daytime
Human comfort index (SSD)	Continuity	Feature Variable	Driver's individual physical feeling index	26.39311
Travel distance	Continuity	Feature Variable	Truck travel distance	2869.54(m)
Atmospheric pressure	Continuity	Feature Variable	Basic meteorological data	884.02(hPa)
Slope of the ramp	Continuity	Feature Variable	Slope angle of the ramp	8(%)
Elevation difference	Continuity	Feature Variable	Elevation difference relative to the loading and unloading position	50(m)
Wind speed	Continuity	Feature Variable	Basic meteorological data	5.1(m/s)
Rainfall	Continuity	Feature Variable	Basic meteorological data	11.03(mm)
Snowfall	Continuity	Feature Variable	Basic meteorological data	10.11(mm)
Travel time	Continuity	Target Variable	Travel time of the truck	56.3(min)

Table 1. Data set structure of the prediction model.

The prediction data of truck travel time has multi-dimensional and multivariable characteristics, there are usually dimensional and order of magnitude differences, and direct application of Bessel formula will adversely affect the training effect of the model. Therefore, Eq. (8) is applied to normalize the sample data of the truck travel time data set and map it to [0,1], so as to ensure that the preference and weight distribution of the data are not affected by the value range of different variables when training the model, improving the prediction accuracy and accelerating the convergence speed.

$$X_{i_d} = \frac{X_{i_d} - X_{min_d}}{X_{max_d} - X_{min_d}} \tag{8}$$

Where, X_{i_d} is the normalized value of the *i* th data in the *d* dimensional data, X_{min_d} and X_{max_d} are the minimum and maximum values of the *d* dimensional data of the input data *X*, respectively.

Evaluation index

Common evaluation indexes include root mean square error, mean absolute error, mean absolute percentage error and determination coefficient. In this paper, Relative Absolute Error (RAE) and Relative Squared Error (RSE) are selected to quantify the prediction accuracy of the truck travel time prediction model. RAE is a commonly used evaluation index (Eq. (9)), which measures the ratio of the absolute error between the actual value and the predicted value to the expected value. The smaller the error between the predicted value and the real value, the smaller the RAE value. RSE is another common evaluation index (Eq. (10)). In the calculation process, the error is squared, and the square of the error will increase rapidly with the increase of the error. Compared with RAE, RSE gives a higher weight to the larger prediction error. The smaller the RSE value, the higher the prediction accuracy and the stronger the adaptability of the model.

$$RAE = \frac{\sum_{i=1}^{n} \mid y_i - \widehat{y_i} \mid}{\sum_{i=1}^{n} \mid \overline{y} - y_i \mid} \in [0, +\infty) \tag{9}$$

$$RSE = \frac{\sum_{i=1}^{n} (y_i - \widehat{y_i})^2}{\frac{1}{n}\sum_{i=1}^{n} (\overline{y_i} - y_i)^2} \in [0, +\infty) \tag{10}$$

Experiment

Experimental environment and parameter settings

The experimental environment of truck travel time prediction in open-pit mine is based on Windows 11 (64-bit) operating system, with a 12th Gen Intel (R) Core (TM) i7-12700 H CPU based on x64 processor, 16 GB memory, 1024 GB hard disk, and a NVIDIA GeForce RTX 3050 Ti GPU. The deep learning framework is PyTorch 1.8 and Python 3.8.

Adam is an optimization algorithm used in deep learning models to replace stochastic gradient descent. It combines the optimal performance of AdaGrad and RMSProp algorithms, with relatively simple hyperparameter tuning, and the default parameters can handle most problems. This method dynamically adjusts the learning rate for each parameter's gradient, avoiding the issues caused by a fixed learning rate. Additionally, by using momentum, it accelerates convergence and avoids local minima and gradient vanishing. In the Adam optimizer, estimates of the gradient and the squared gradient are calculated using exponentially decaying averages. Since the estimated mean and variance have some bias in the early stages, Adam introduces a bias correction mechanism. These corrections avoid biases during the early training phase, especially for smaller mini-batch sizes and early-stage training, making the estimated gradient and second-order moments more accurate. Due to its performance in handling complex models and large-scale data, in the model training process, the Adam optimization method is used to update the LSTM-TabTransformer parameters, and the cosine annealing method is used to update the learning rate. The feature vector dimension dim of the TabTransformer embedding layer is set to 32, the depth is 6, the number of attention heads in the multi-head attention mechanism is 8, and a total of 50 epoch training is performed (Table 2).

Embedding layer feature vector dimension (dim = 32)

The dimension of the embedding layer determines the representational capacity of each categorical feature when it is transformed into a vector. A smaller dimension may not capture enough information, while a larger dimension would lead to overfitting and increase computational burden. In TabTransformer, a dimension of 32 is a reasonable value as it ensures that the vector representations of categorical features have sufficient expressive power, and without wasting excessive computational resource.

Parameter name	Parameter value	Parameter name	Parameter value
Number Continuous	9	Data_Dropout	0.01
Dim	32	Optimizer	Adam
Heads	8	Number Epoch	50
Depth	6	MLP_act	ReLU

Table 2. LSTM-TabTransformer parameters setting.

Index value	LSTM-TabTransformer	LSTM	BERT	Transformer
RSE	**1.806**	13.878	6.612	59.507
RAE	**0.176**	1.257	0.559	5.678

Table 3. RSE and RAE values of the comparison models.

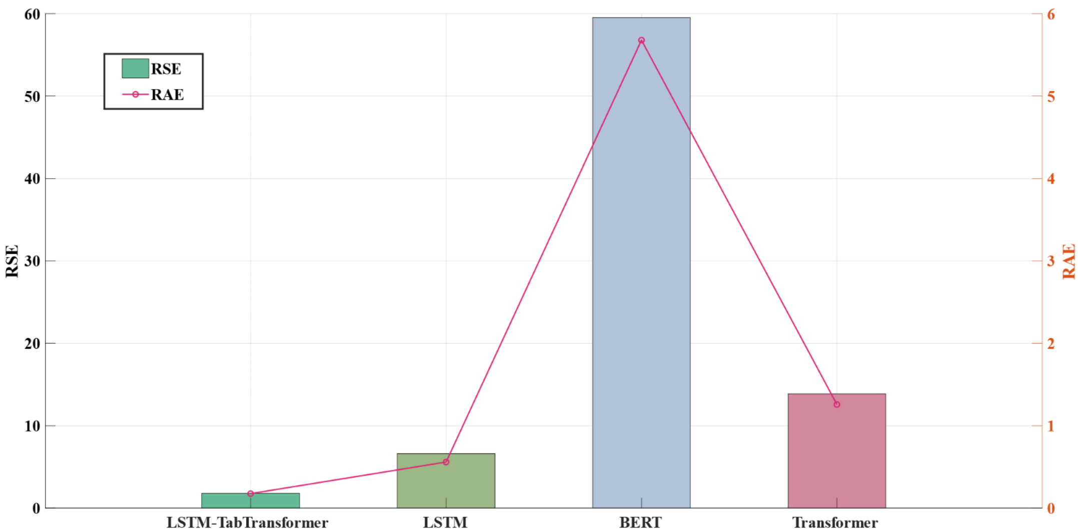

Fig. 9. RSE and RAE histogram of the comparison models.

Depth (depth = 6)
A depth of 6 layers performs well in many tasks. Increasing the depth will enhance the model's expressive power, but it also increases the computational complexity. For TabTransformer, depth of 6 is sufficient to handle the more complex nonlinear relationships in most cases.

Number of attention heads in multi-head attention mechanism (8 heads)
In many NLP and tabular data tasks, 8 heads provide an excellent trade-off between performance and computational efficiency.

Theoretical and experimental basis
In TabTransformer, the value of embedding dimension and depth offer satisfactory performance across many tasks. For example, in some tabular data tasks, an embedding dimension of 32, a depth of 6 layers, and 8 attention heads have shown great generalization ability. Embedding Dimension: 32 is a reasonable middle value, effectively representing categorical features without causing excessive computational cost. Model Depth: A depth of 6 layers typically provides enough expressive power without being overly complex, making it suitable for most tabular data tasks. Number of Attention Heads: 8 heads effectively capture diverse relationships between features, while the computational cost remains moderate.

Comparative analysis
LSTM, Transformer and BERT[21] are selected to compare the prediction results of truck travel time in open-pit mines with LSTM-TabTransformer. The initialization parameters of the Transformer and BERT models : the embedding dimension is 512, the number of multi-head attention mechanisms is 8, the attention mechanism dimension is 64, and the ReLU activation function is selected. The initialization parameters of the LSTM model : the dimension of the hidden state is 32, and the number of model layers is 8. The training epoch of the above three truck travel time prediction models is 50. The comparison of the evaluation indexes of the experimental prediction results is shown in Table 3, and Fig. 9 is the comparison histogram of the prediction result.

It can be seen from Table 3; Fig. 9 that the prediction effect of LSTM-TabTransformer model on truck travel time in open-pit mines is better than that of Transformer, BERT and LSTM. The smaller RSE and RAE values indicate that the model has better performance in this task. The traditional Transformer, LSTM and BERT perform poorly, indicating that for the open-pit mine truck travel time prediction with large randomness and complex data composition, the traditional single model cannot well capture the characteristics and variation

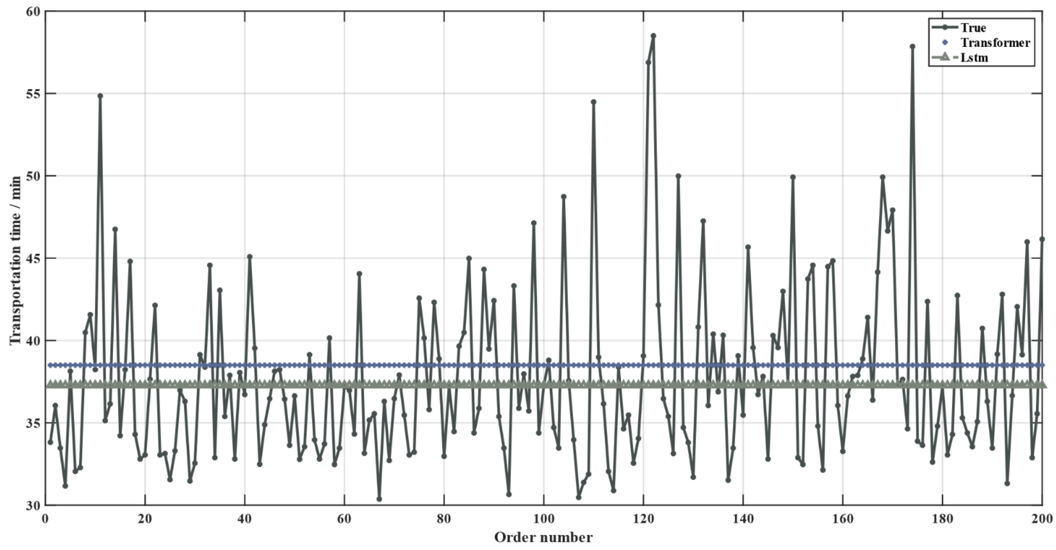

Fig. 10. Prediction results of LSTM and Transformer.

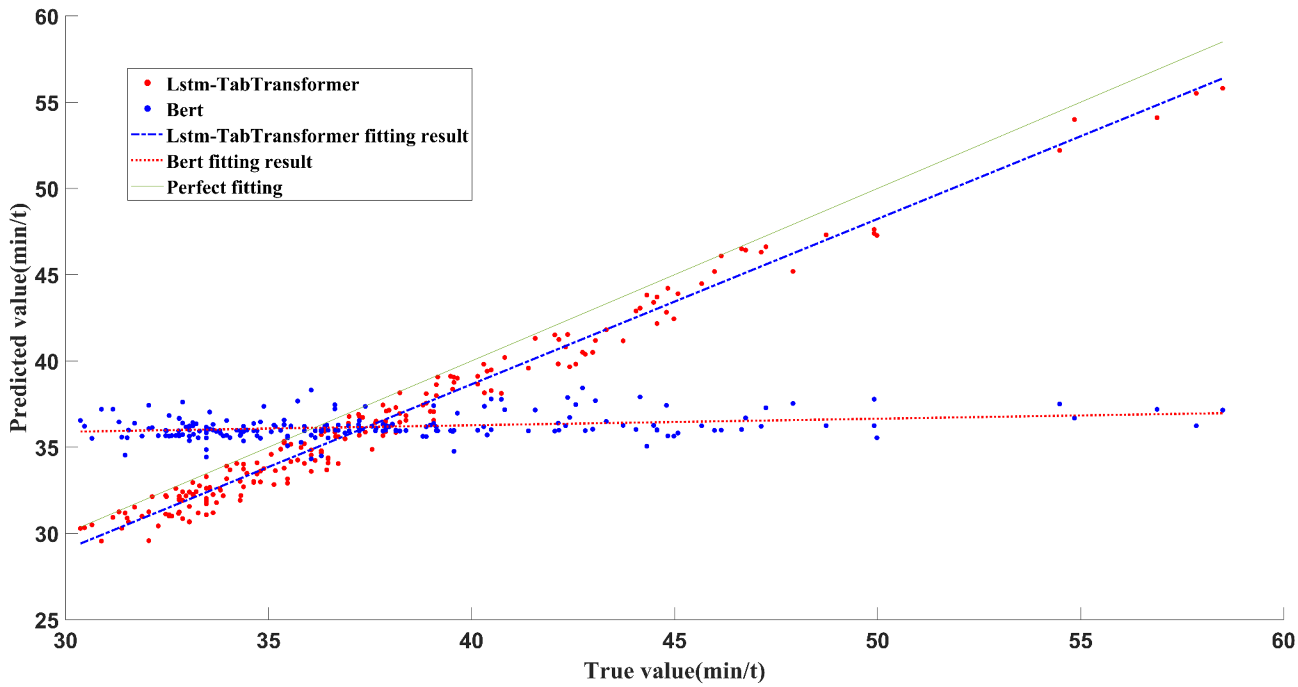

Fig. 11. Scatter plot and fitting curves.

rule of the data, resulting in unsatisfactory prediction effect. The combined model has obvious advantage than the single model.

Randomly sampling 200 test data from 2000 test set data to verify and evaluate the prediction performance of the model. The prediction results of LSTM and Transformer models are shown in Fig. 10. Figure 11 is the scatter plot of the prediction results and fitting curves of LSTM-TabTransformer and BERT. Figure 12 is the prediction results of LSTM-TabTransformer and BERT.

It can be seen from Fig. 10 that for the multi-factor and multi-dimensional open-pit mine truck travel time prediction, the results of the traditional LSTM and Transformer models gradually tend to be average, indicating that the traditional model is not applicable with multiple types of data in the input layer.

From Fig. 11, it can be seen that the linear fitting linear slope of the BERT model simulation result is 0.04, which represents the model's fitting difference for the linear relationship between input and output, and the

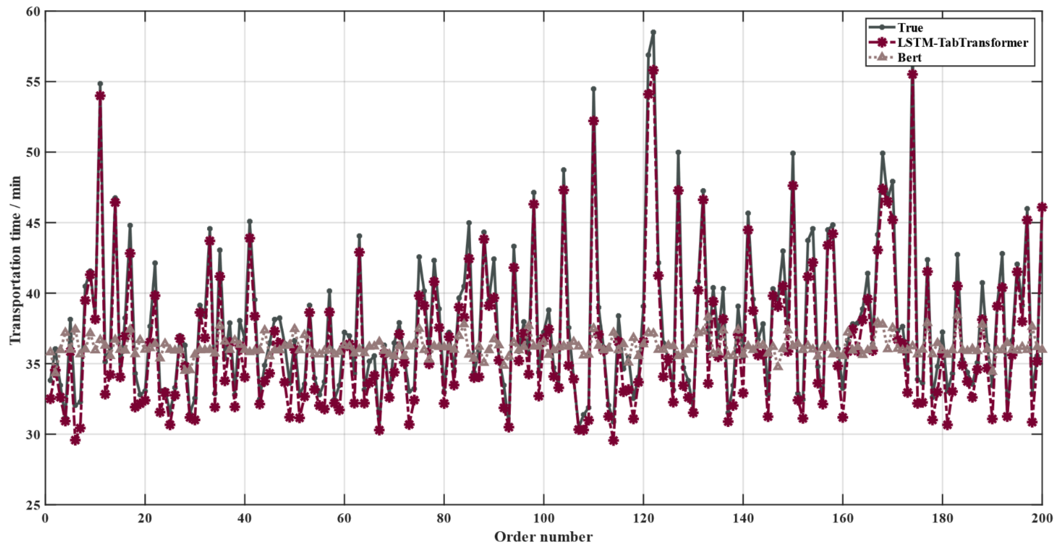

Fig. 12. Prediction results of BERT and LSTM-TabTransformer.

prediction accuracy is low. The linear fitting slope of the LSTM-TabTransformer model simulation results proposed in this paper is 0.96, which means that the LSTM-TabTransformer model proposed in this paper fits the nonlinear relationship between features and outputs well, and can better capture the potential relationship between data, thus having the higher prediction accuracy.

The BERT model performs poorly in time prediction results, and the predicted value is usually close to 36 or even equal to 36. The reason is that the BERT model is difficult to find the implicit relationship in the prediction data of open-pit mine truck travel time. The model is not sensitive to extreme values and fluctuations, and can not accurately capture the rule of extreme values and variationss. In contrast, the LSTM-TabTransformer model has better prediction fit and minimum prediction error, and is more applicable to the prediction of truck travel time in open-pit mines.

It can be seen from Fig. 12 that the error between the predicted value and the real value of the BERT model is large, mainly because the traditional model is not suitable for the complex time series and multi-factor tabular data, and it is unable to deeply explore the implied variation rule of truck travel time under the influence of multiple factors. The LSTM-TabTransformer model proposed in this paper is a combined machine learning model designed for complex time-series and multi-factors tabular data. It can use the self-attention mechanism of the Transformer model to obtain the potential variation rules of feature data. It can also focus on all positions in the feature category sequence, better capture global information, and make the model more comprehensively understand the feature sequence. In addition, the proposed LSTM-TabTransformer model uses the LSTM gate mechanism to realize the feature capture of long-term time-series data dependencies, making the predicted value close to the real value, with the high prediction accuracy and excellent generalization ability.

Conclusion

A combined machine learning model of LSTM-TabTransformer for truck travel time prediction in open-pit mines is proposed. LSTM and TabTransformer are combined to learn the implicit logic of continuous data and category data in the shunted truck travel table data of open-pit mines. By concatenating the learned data features, the prediction results are determined after MLP dimensionality reduction and full connection.

The LSTM-TabTransformer model is based on the self-attention mechanism of the TabTransformer and the gating mechanism of the LSTM. It deeply learns the dependencies between time series and category features data, and overcomes the problem that a single machine learning model is difficult to accurately capture the composite features of time series sequences.

Based on the truck travel time prediction training data set collected from the exampel open-pit coal mine, the truck travel time prediction experiment was carried out. The results show that compared with the traditional machine learning models such as Transformer, BERT, and LSTM, the prediction results of the LSTM-TabTransformer model proposed in this paper have RSE of 1.806 and RAE of 0.176. The prediction accuracy of the LSTM-TabTransformer is higher and the performance is better. It can be applied to the real-time optimal scheduling decision of the truck in open pit mines, and can also be used for other similar problems.

Data availability

All data supporting the findings of this study are available on request from the authors.

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Author contributions

JZ: conceptualization, methodology, writing-review & editing; SR: methodology, data curation, coding, writing-original draft; LG: data curation, coding, review & editing.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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